



SQ TECH SOLVER

IT INNOVATION EXHIBITION 30 DECEMBER 2025

95 FINAL PROJECT Presentation

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Supervised by

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Project Overview

Secure QR Code Generator System

The Secure QR Code Generator System is a centralized web-based platform that integrates QR generation, security enforcement, and monitoring into a single solution. Users can generate, manage, share, and analyze QR codes while the system transparently enforces security policies at every stage, from creation to scanning and content delivery.



Problem Statement



Problem Statement

Traditional QR codes provide unrestricted access once scanned, exposing users to fraud risks such as phishing, malware distribution, credential theft, and unauthorized data access. Existing QR solutions prioritize convenience over security and lack encryption, access control, and monitoring, making QR codes an increasingly exploited attack vector in modern digital environments.

Objective



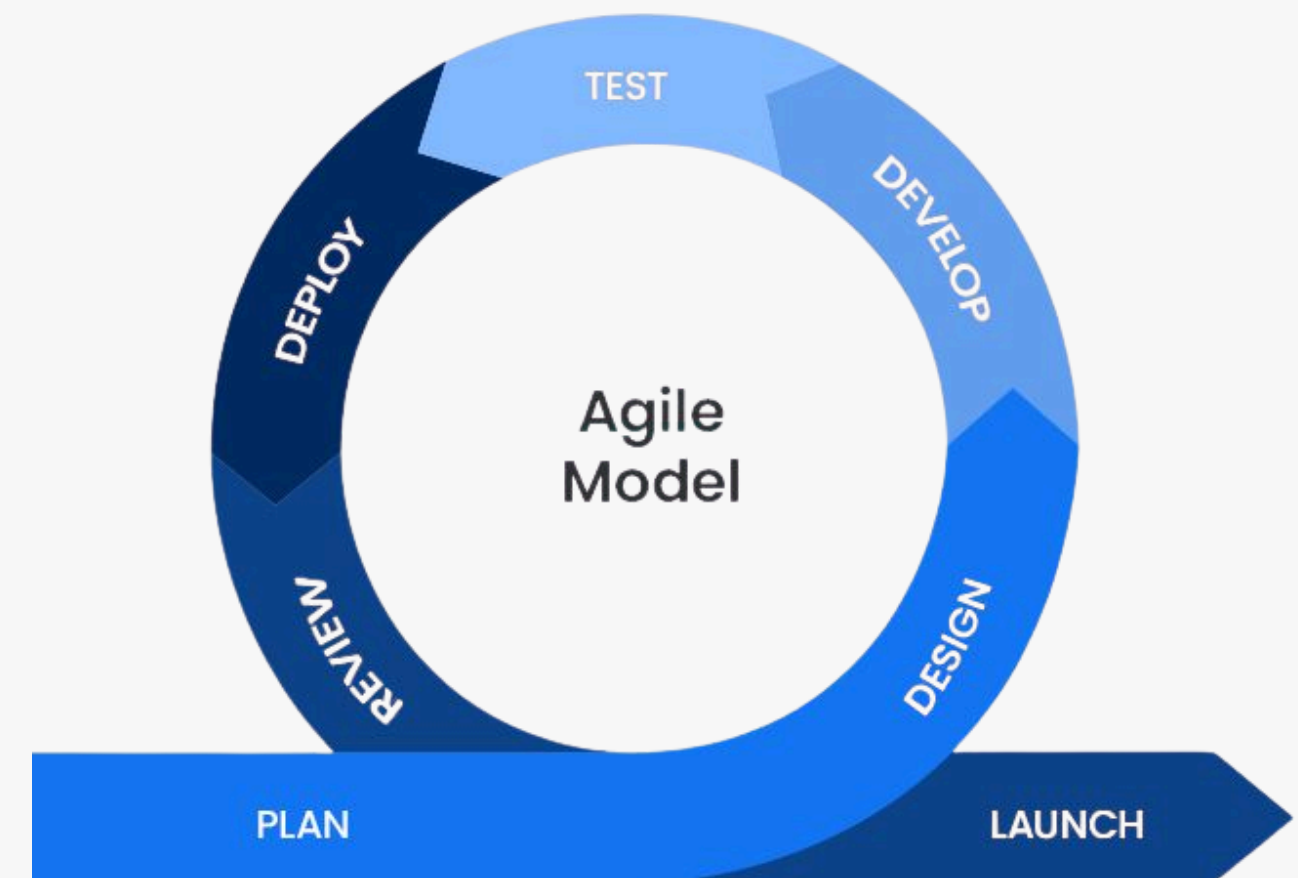
Objective

- To develop a secure, centralized QR code system with strong encryption
- To prevent QR-based fraud through malware scanning and content validation
- To enforce controlled access using password and OTP verification
- To provide monitoring, logging, and auditability of QR usage
- To balance high security with user-friendly system design

Methodology

Methodology

- The project follows the Software Development Life Cycle (SDLC) using the Agile development model
- Agile is selected to support iterative development, continuous testing, and security refinement
- Development is conducted in multiple sprints, each delivering a functional system module
- Each sprint includes requirements analysis, system design, implementation, testing, and evaluation
- Continuous feedback is used to identify vulnerabilities, improve usability, and enhance system performance
- The Agile approach ensures flexibility, rapid improvement, and strong security enforcement throughout development



Journey

Phase 1 : Plan
14-23 April

Phase 2 : Design I
12-26 May

Phase 2 : Design II
9-14 June

Phase 3 :
Development I
25-30 June

Phase 3 :
Development II
27 July-29 Oct

Phase 3 :
Development III
1 Nov-10 Dec

Phase 4 : Testing
11-25 Dec

Phase 5 :
Maintenance
25-27 Dec

Phase 6 : Review
27-30 Dec

Function

Main Function + Optional

Login

Generate QR Code

Manage QR Code

Register Account

Scan QR Code

Share QR Code

Forgot Password

Analyze QR Code

Educate Yourself

Make Friend

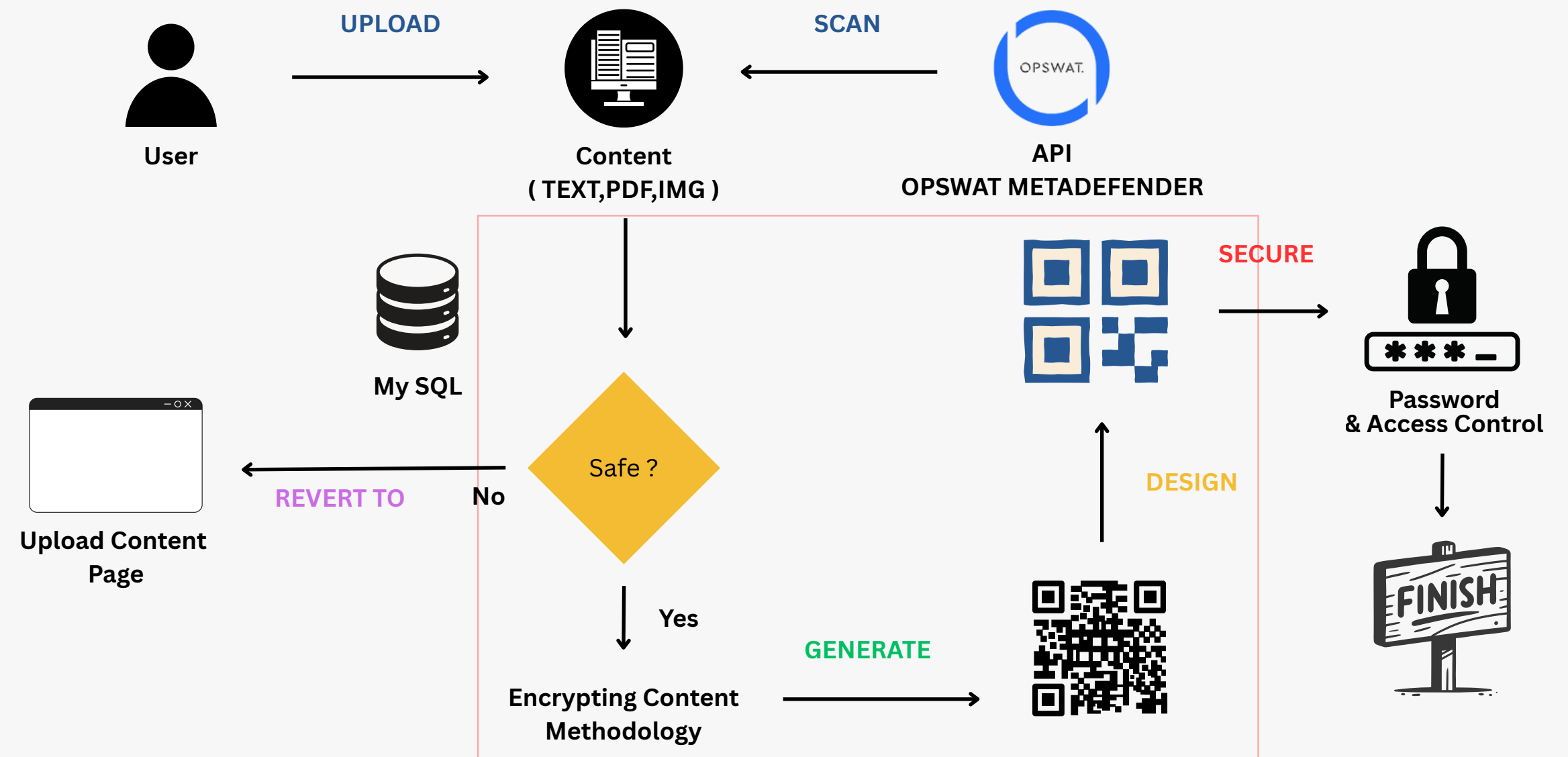
View Profile

Track Scan QR Code

Logic

Logic + Security + Main Function

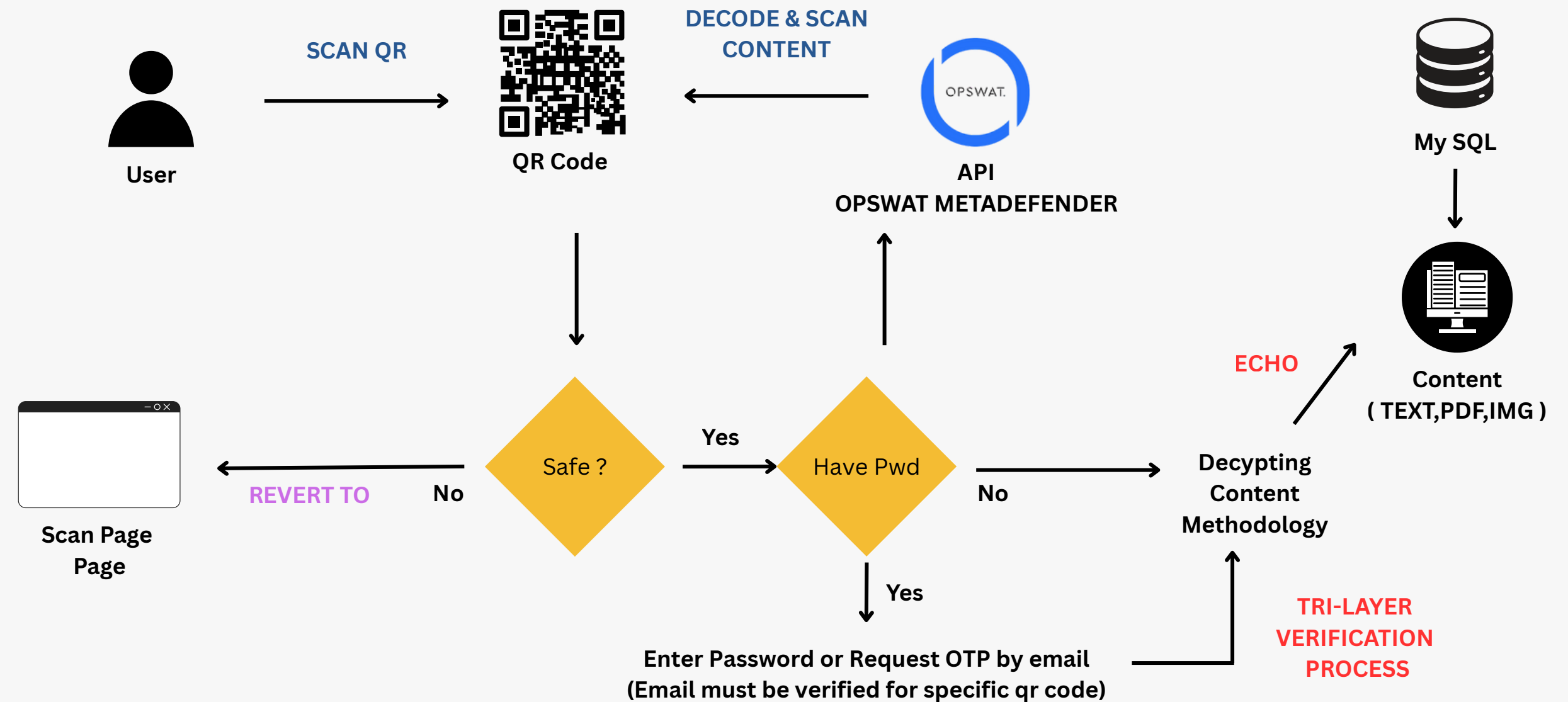
Generate QR Code



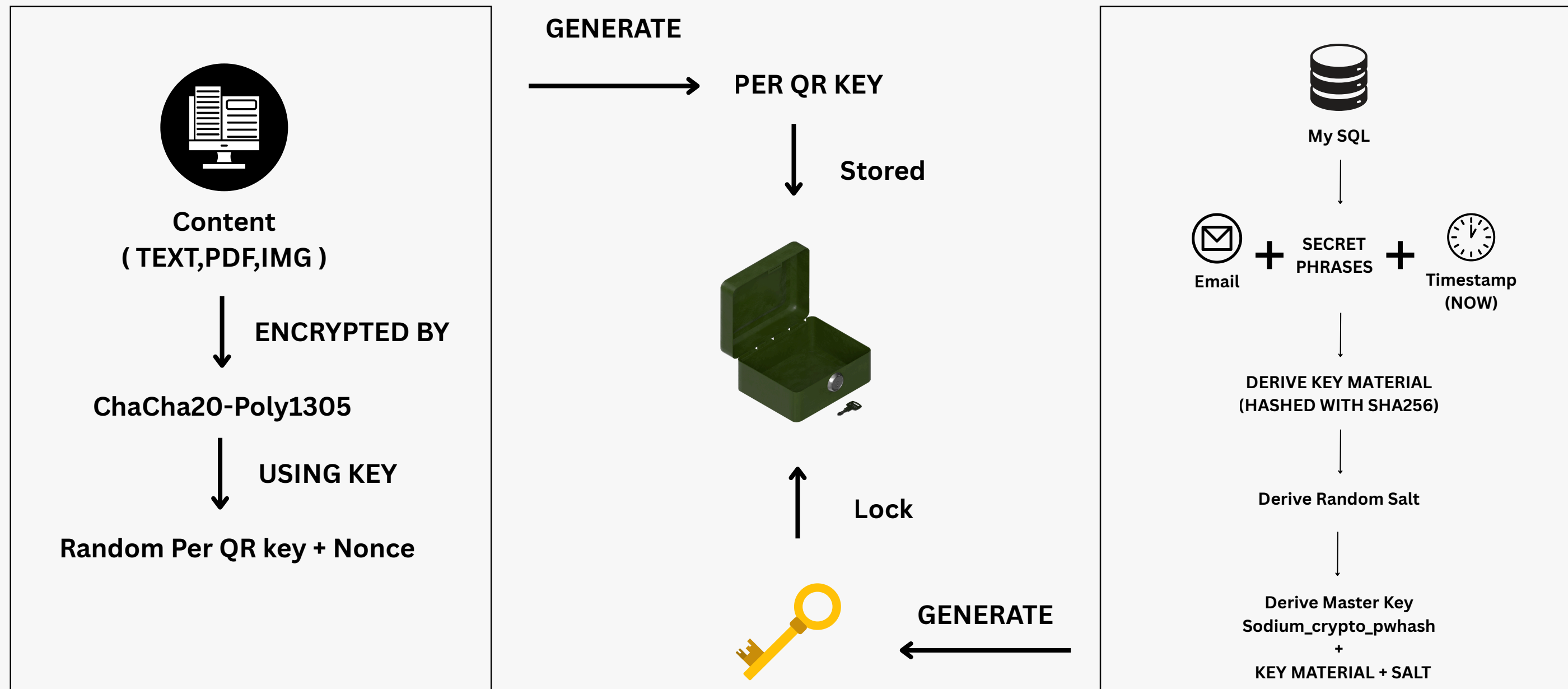
Logic

Logic + Security + Main Function

Scan QR Code

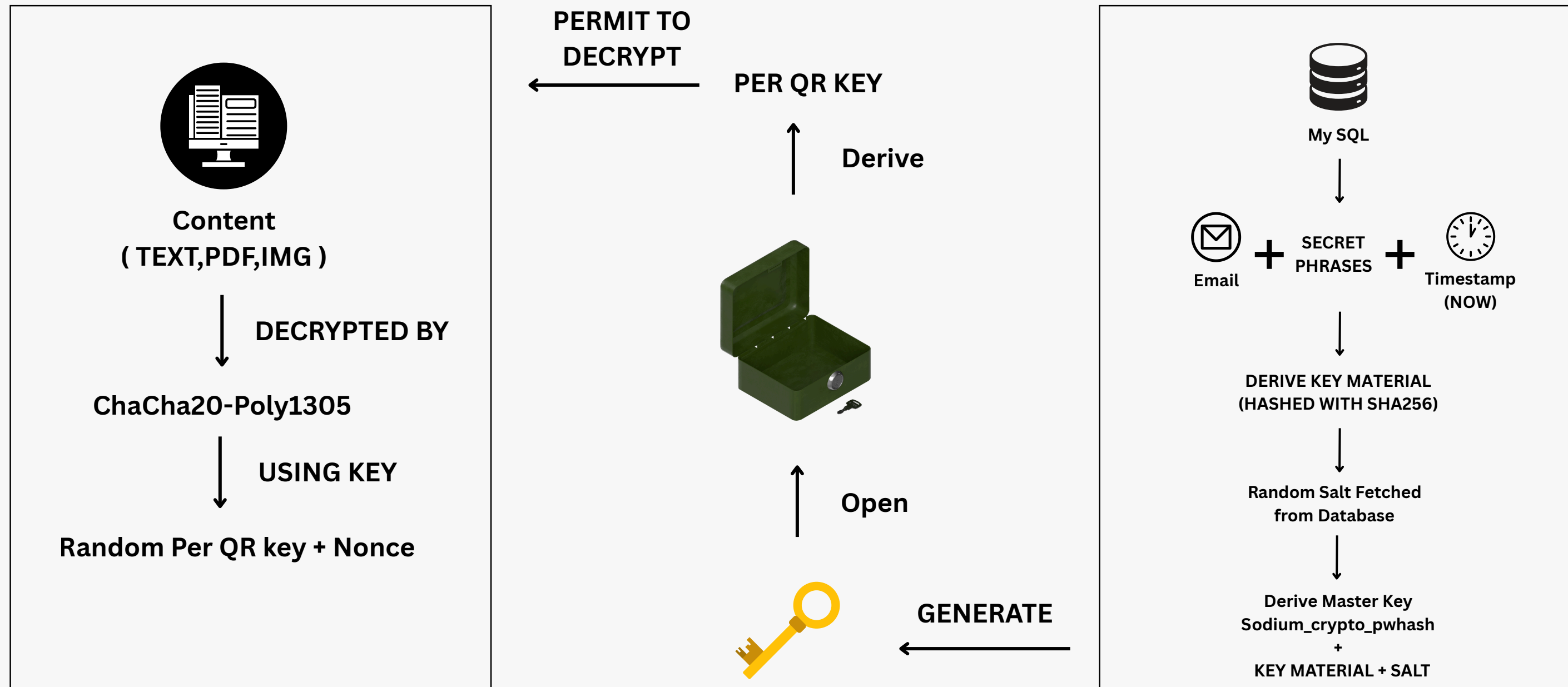


Encryption Algorithm



Per Qr Key will be encrypted with nonce-key for serial number and user key (master key). The concept stored in the box can be defined as envelope encryption

Decryption Algorithm



Per Qr Key will be encrypted with nonce-key for serial number and user key (Master key). The concept stored in the box can be defined as envelope encryption

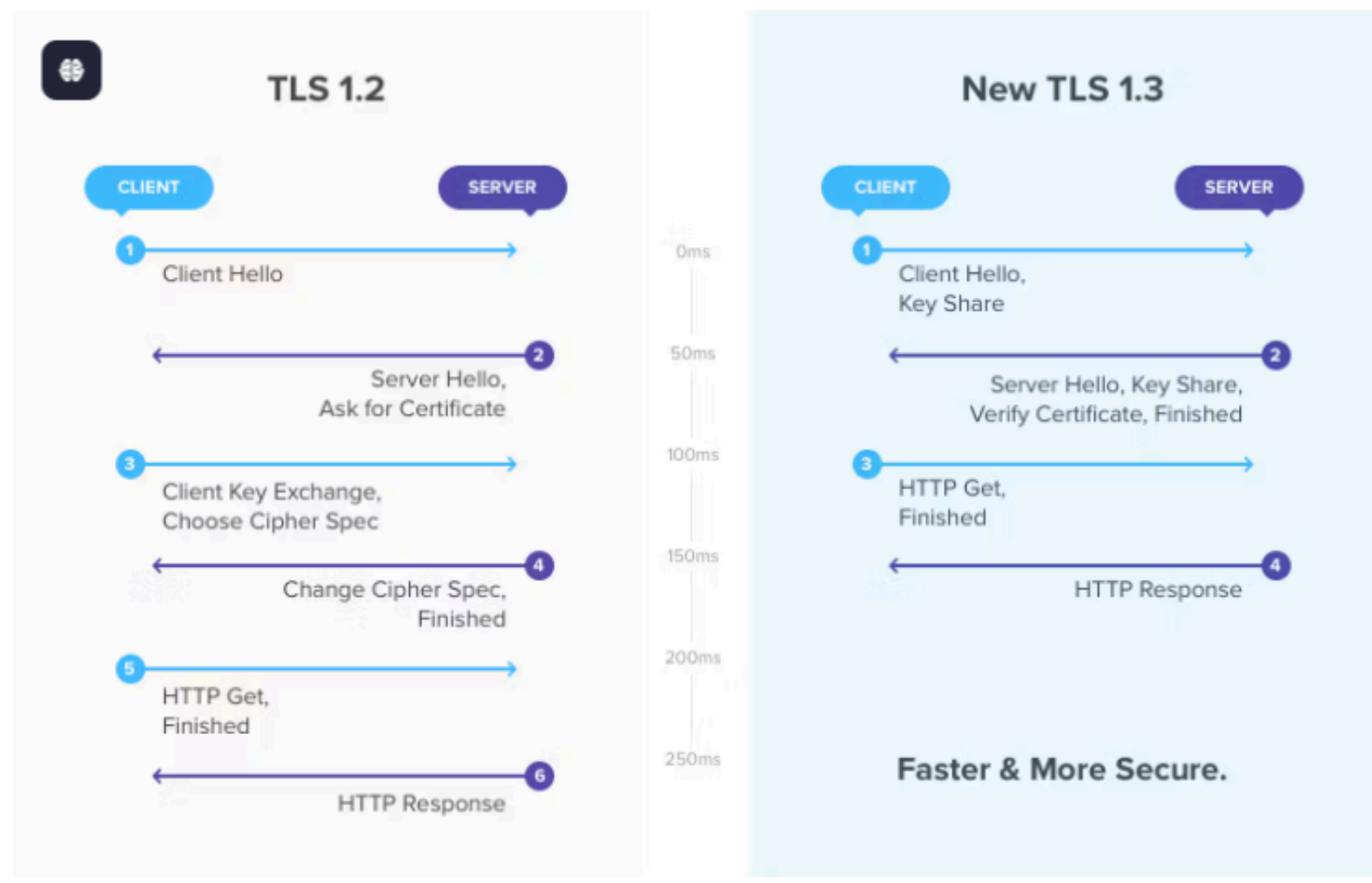
Why ?

Category	Algorithm/Method	Compared Algorithms	Technical Justification	ISO/IEC 27001 Alignment
Encryption Algorithm	ChaCha20-Poly1305	AES-GCM, AES-CBC	ChaCha20-Poly1305 is a modern AEAD cipher resistant to timing/cache attacks. Poly1305 ensures message integrity and detects tampering. Unlike AES-CBC, it avoids padding oracle attacks, and unlike AES-GCM, it does not require hardware acceleration to remain secure and performant. Its simpler design reduces implementation errors, making it highly reliable for QR content protection.	A.10.1 Cryptographic controls
Password Key Derivation Function (PKDF)	Argon2id	PBKDF2, scrypt	Argon2id is memory-hard, resisting GPU/ASIC brute-force attacks, and combines side-channel and trade-off attack resistance. PBKDF2 and scrypt rely more on CPU cycles, making them vulnerable to modern parallel attacks. Argon2id is optimal for securely deriving keys for QR content encryption.	A.10.1 Cryptographic controls
Transport Layer Security (TLS)	TLS 1.3	TLS 1.2, SSL 3.0	TLS 1.3 removes outdated primitives, enforces forward secrecy, reduces handshake latency, and prevents downgrade attacks. TLS 1.2 and SSL 3.0 are vulnerable to known attacks (e.g., POODLE, BEAST). TLS 1.3 ensures secure, authenticated QR data transmission.	A.13.2 Information transfer
Access Control	Role-Based Access Control (RBAC)	DAC, MAC	RBAC enforces least privilege by assigning permissions to roles, reducing insider misuse. DAC (Discretionary) and MAC (Mandatory) provide less flexible or scalable access control for multi-user QR systems. RBAC ensures users only access QR codes they are authorized for.	A.9.1 Business requirements for access control

Why ?

TLS 1.3 with ChaCha20-Poly1305 (AEAD)

- TLS 1.3 (Transport Layer Security) is the latest cryptographic protocol that encrypts data between client and server to prevent eavesdropping and tampering. It provides faster handshakes, reduced latency with 0-RTT (early data), and stronger security by supporting only modern ciphers with Perfect Forward Secrecy.
- ChaCha20-Poly1305 is an AEAD (Authenticated Encryption with Associated Data) cipher used in TLS 1.3. ChaCha20 provides fast, stream-based encryption resistant to timing attacks, while Poly1305 ensures integrity and authenticity, detecting any tampering. Together, TLS 1.3 with ChaCha20-Poly1305 ensures HTTPS data is confidential, authenticated, and tamper-proof, even on devices without AES hardware acceleration.



Why ?

P-H-C/phc-winner-argon2

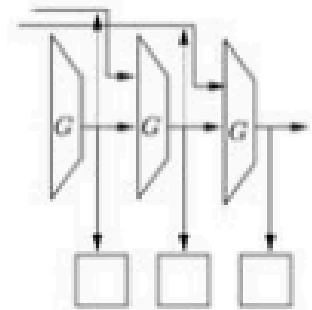
The password hash Argon2, winner of PHC

68
Contributors

2
Used by

5k
Stars

446
Forks



P-H-C/phc-winner-argon2: The password hash Argon2, winner of PHC

The password hash Argon2, winner of PHC . Contribute to P-H-C/phc-winner-argon2 development by creating an account on GitHub.

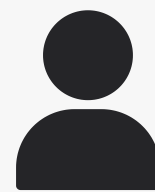
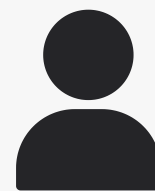


- **Memory-Hard and GPU-Resistant** – Requires significant memory for computation, making brute-force attacks using GPUs, FPGAs, or ASICs much more expensive than PBKDF2.
- **Side-Channel Resistant** – Combines data-dependent and data-independent memory access patterns to protect against cache-timing and other side-channel attacks.
- **Hybrid and Modern Design** – Argon2id merges Argon2d and Argon2i benefits, uses the fast and secure BLAKE2b hash internally, and is standardized in RFC 9106.
- **Tunable Parameters for Security vs Performance** – Adjustable memory cost, time cost, and parallelism allow flexible deployment while ensuring strong protection against modern attacks.

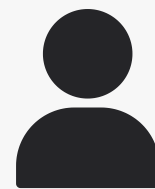
Logic

Logic + Security + Main Function

Share QR Code



DOWNLOAD



Logic

Logic + Security + Main Function

Tri-Layer Verification

Tri layer verification is implemented for qr code with psswrd and no psswrd.

1. Decode the qr code.
2. Fetch the token from the qr code and compared it with token within db.
3. After that, get the qr code id.
4. If qr code have pwd, then compared the hash password with hash password within the db where the qr code id is similar with the token.

This verification is conducted after API has ensured the content of the qr code is safe to be access in db.

Logic

Verify QR Code

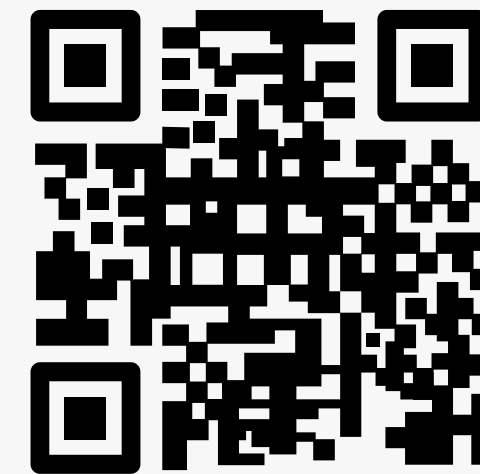
- Owner must choose to verify the QR Code for one time access or life time access.
- If one time access : Recipient can only access QR code through email for first time only, then they need to ask owner to verify it again and vice versa

Access QR Code



With Password

- User can access the pw if they have the password
- They also can access it with request otp



Without Password

- User can access freely
- Password still exist but the system will define it.

Security

Chacha-20Poly1305

Argon2id

Password bcrypt

Access Verification

Password

Access Tracking

One Time Passcode

Separated Content

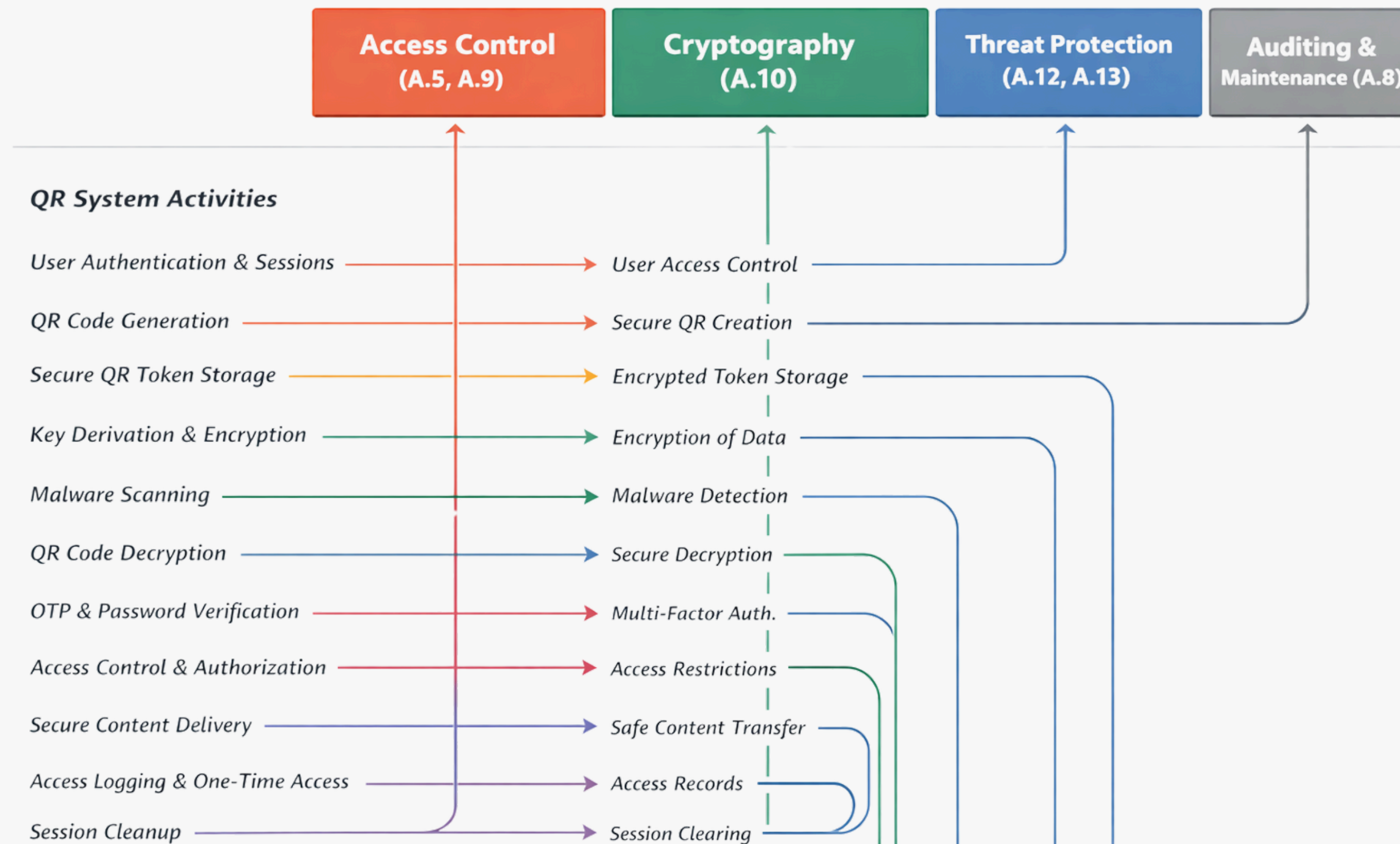
Tri-Layer verification

Permission Control

Malware Verification

IP Address Tracking

ISO 27001



QR System Activities vs ISO/IEC 27001:2022 Mapping

SAST TEST

The screenshot displays the Snyk Code Analysis interface for a project named "Secure-QR-Code-Generator-Integrate-With-API". The interface is divided into a left sidebar, a top header, and a main content area.

Left Sidebar: Contains the Snyk logo, organization name "danielhensem", and navigation links for Dashboard, Projects (selected), Integrations, Members, and Settings.

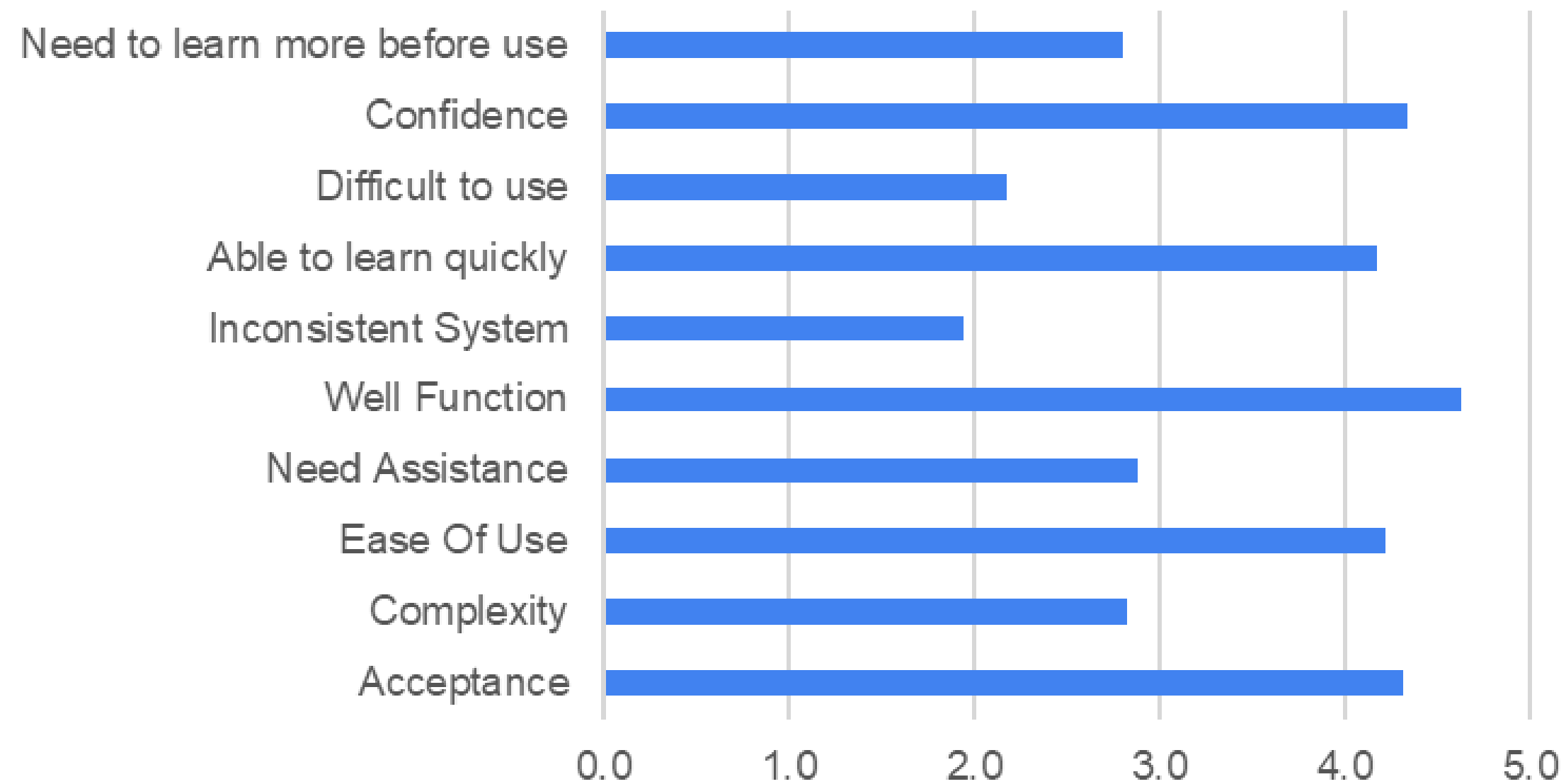
Top Header: Shows the project name "danielhensem/Secure-QR-Code-Generator-Integrate-With-API" and a "main" branch selector. It also includes a search bar, a star icon, a user profile icon, and a "Chat" button.

Main Content Area:

- Code Analysis Overview:** Displays the project name, a "Code Analysis" icon, and tabs for Overview, History, and Settings. It shows the creation date "Mon 3rd Nov 2025" and a snapshot for commit "fdf7f41" taken by a recurring test 5 hours ago. A "Retest now" button is available.
- Metadata:** Includes sections for "Imported by" (m_daniel_haikal), "Project Owner" (m_daniel_haikal), "Environment" (Frontend, Backend, External, Mobile), and "Business Criticality" (Medium).
- Lifecycle:** Shows "Production" and "Development" environments.
- Analysis Summary:** Indicates "424 analyzed files (28%)" and a "Repo breakdown" link.
- Issues:** A section titled "Issues" with a count of 34. It includes a search bar and filters for Severity (High, Medium, Low) and Priority Score (0-1000). The status filter shows 34 Open issues and 0 Ignored issues.
- Vulnerability Details:** A specific issue is highlighted: "Path Traversal" (SNYK CODE | CWE-23). It shows a code snippet from "TCPDF-main/TCPDF-main/include/tcpdf_static.php" with a line number of 1973. The vulnerability description states: "Unsanitized input from an HTTP header flows into file_get_contents, where it is used as a path. This may result in a Path Traversal vulnerability and allow an attacker to read arbitrary files." The score for this issue is 815.

USABILITY TEST

SUS Form Analysis

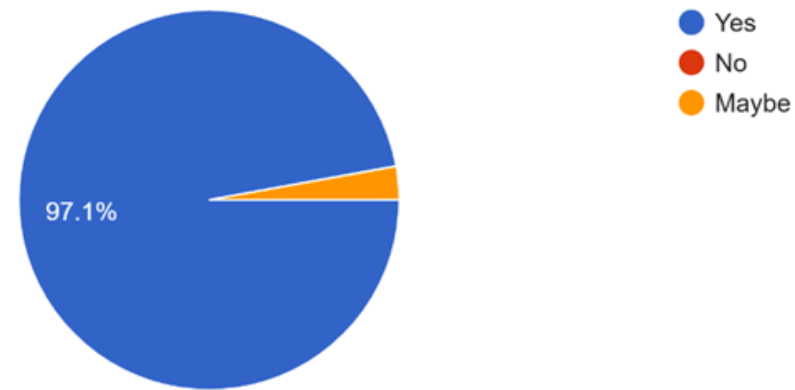


SUS Score

72.64

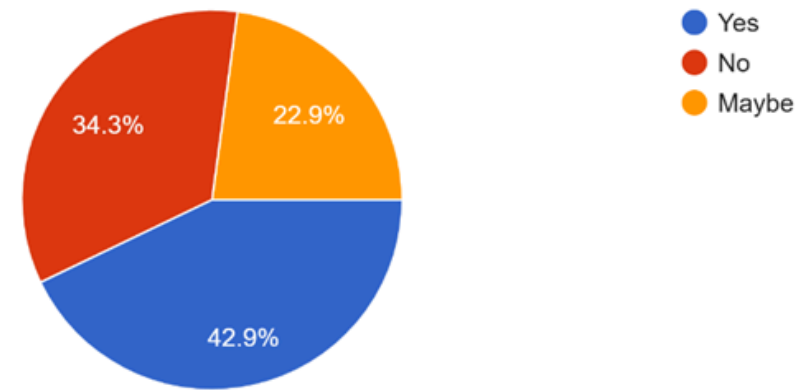
1. Have you ever used a QR code before?

35 responses



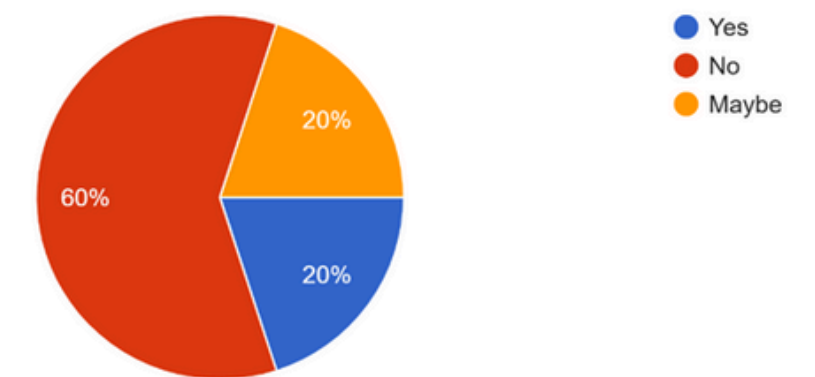
2. Have you ever heard of QR code fraud cases?

35 responses



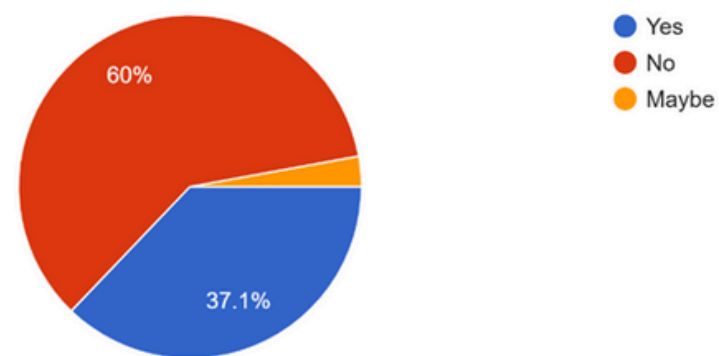
3. Do you think QR codes are safe to use for sharing sensitive information?

35 responses



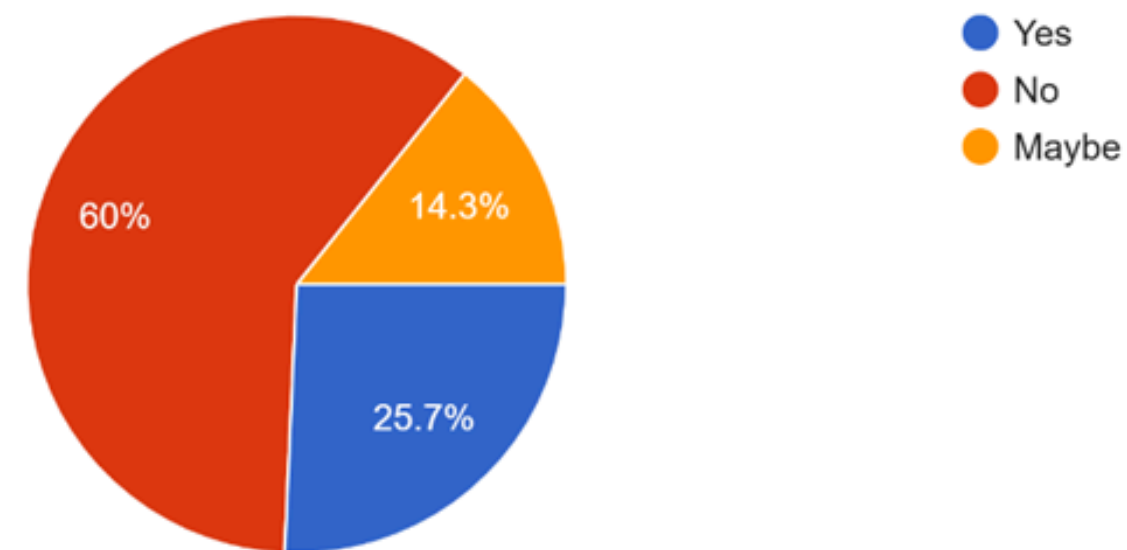
4. Have you ever used a QR code system that requires an OTP/password and includes multiple layers of data protection?

35 responses

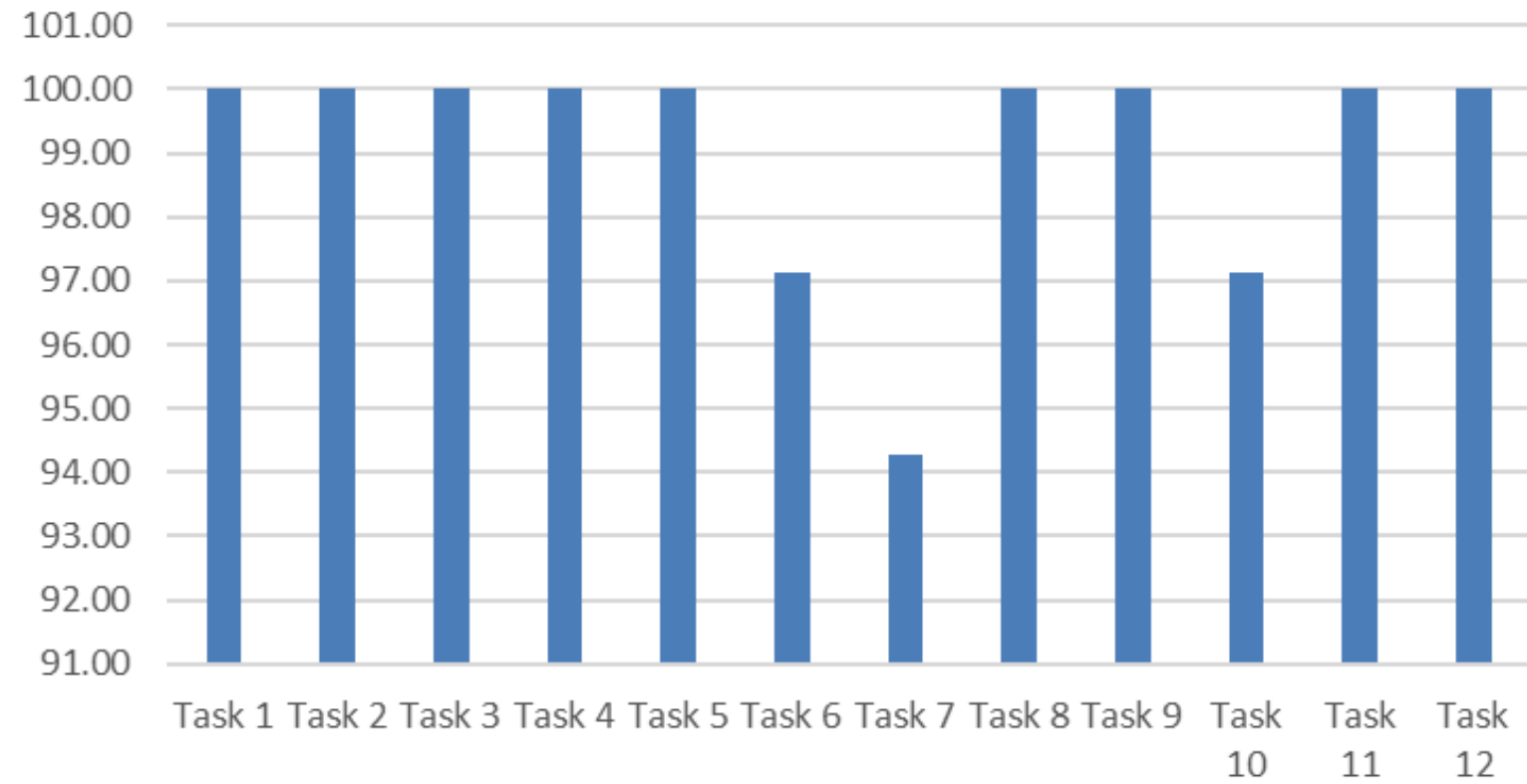


5. Have you ever used a QR code Generator system that managing your QR Code systematically?

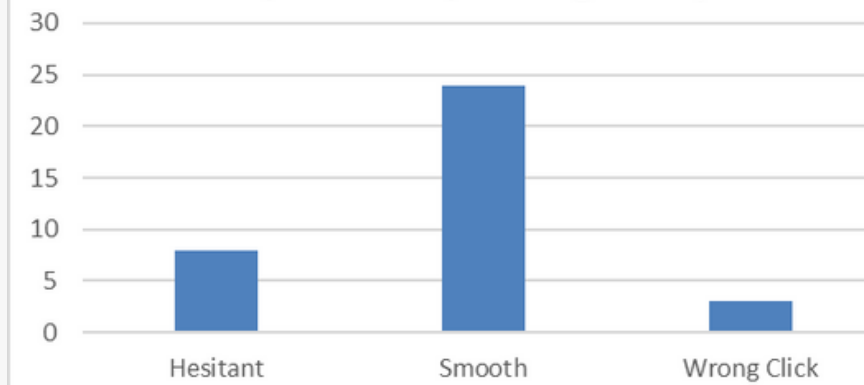
35 responses



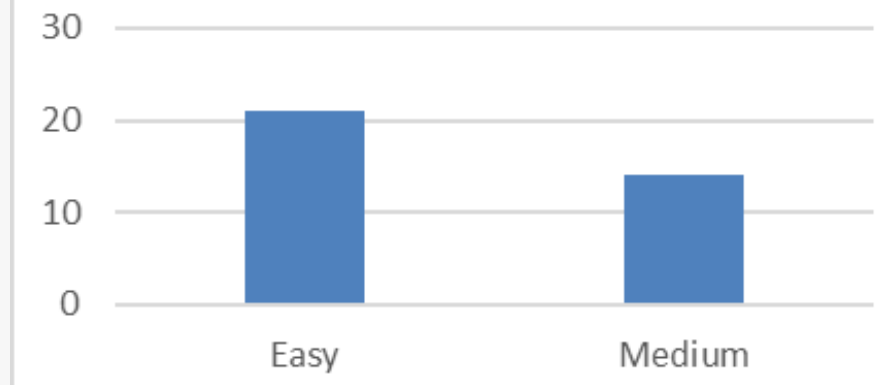
Completion Task



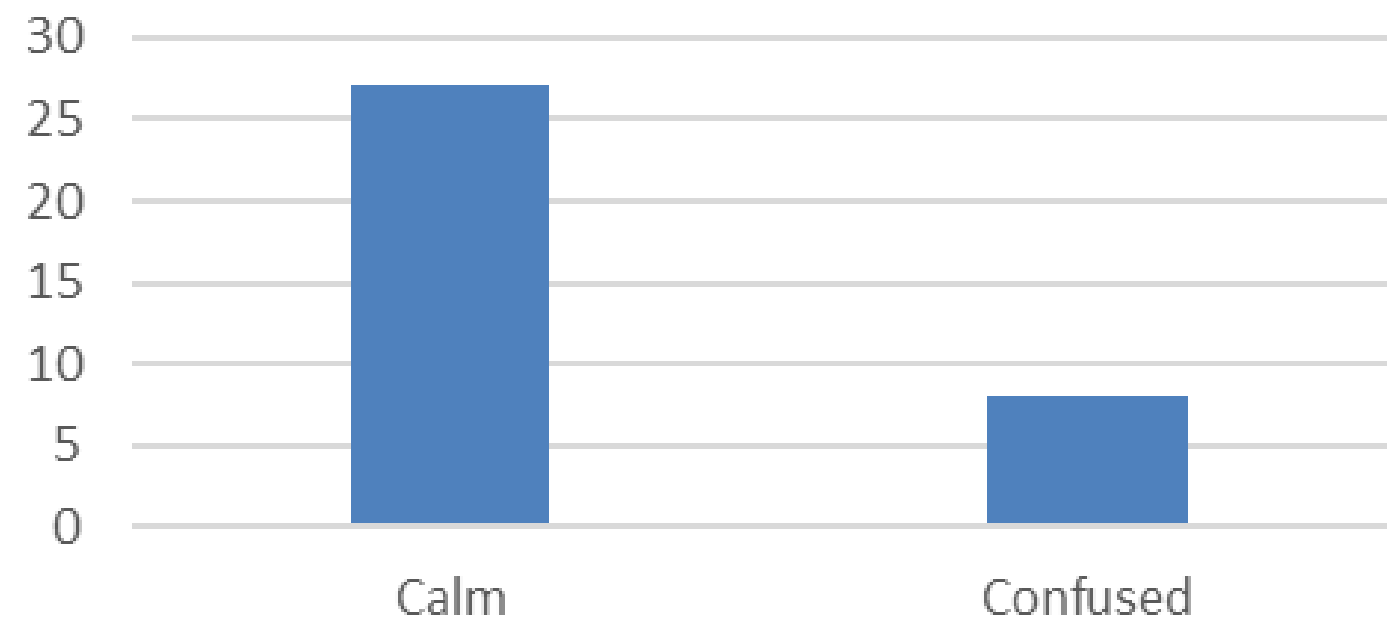
Count of Navigation behaviour (Smooth / Hesitant / Wrong clicks):



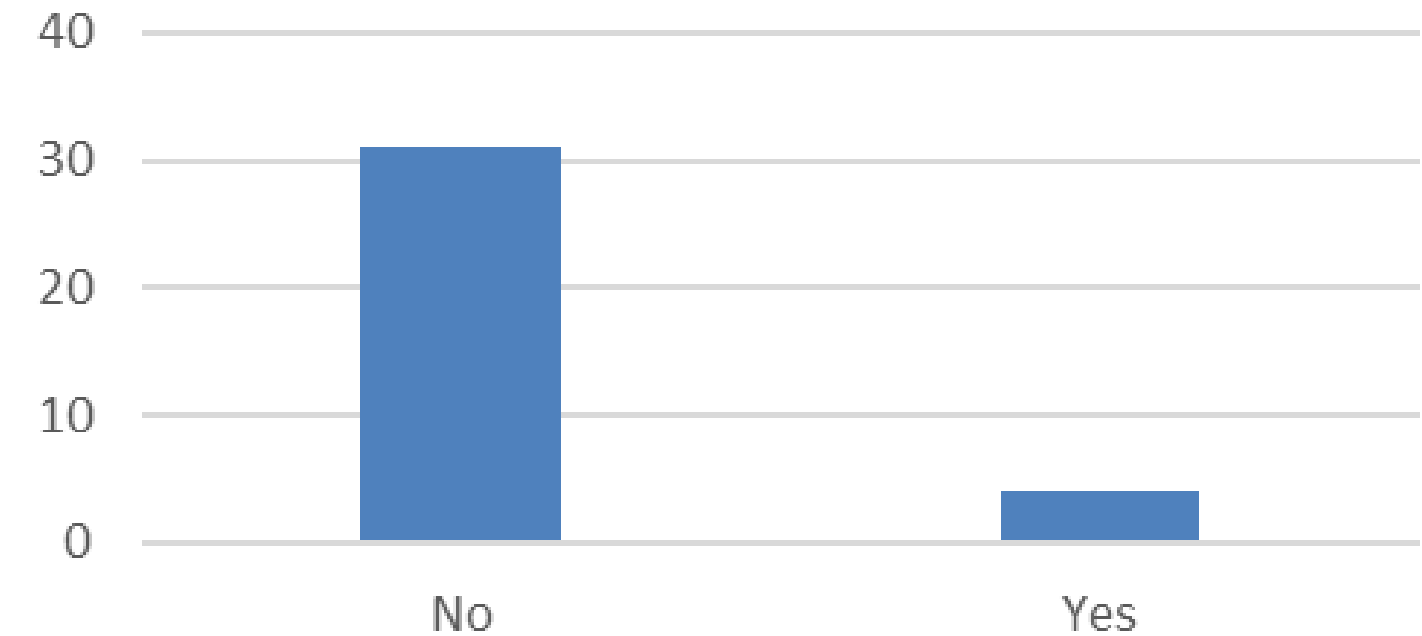
Count of Was the user able to locate buttons easily?



Count of User reaction to errors (Calm / Panicked / Confused):



Count of Did user retry or ask for help immediately?



Nice project

The system performs well

**Overall, the system is reliable and
convenient to use**

The best project qr code management
system

**i love this project and it is suitable for all
users as long as they got good guidance**

Everything goodd i love this project

I think this system will be really secure

**It's interesting to me because this is the first
time i saw a system that can do all the
functions for QR**

Good project

Type Attack

Attack Scenarios in Traditional QR Systems

QR Code Tampering

Session Hijacking

Unauthorized Access

Man-in-the-Middle
(MITM) Attacks

Data Leakage

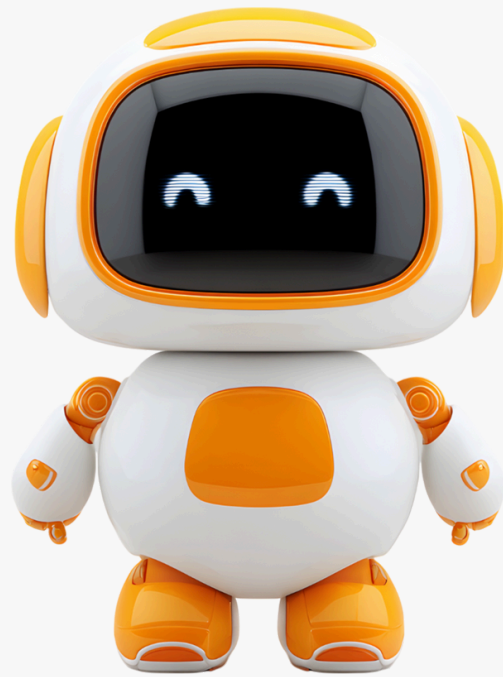
Replay Attacks

Industry Practise

Industry Alignment & System Differentiation

- **Designed using industry-grade envelope encryption**
- **Uses modern AEAD cryptography and memory-hard KDFs**
- **Enforces Zero Trust access to QR-linked content**
- **Integrates malware scanning prior to encryption and storage**
- **Provides audit logging aligned with ISO/IEC 27001**

Interesting Feature



AI Chatbox



**Real Time
Tracking**



**QR Code Trend
Analysis**

QR Code Simulation

**Restaurant Booking
Simulation**

Book Purchase

**Share Confidential
Information (pdf)**

Event Registration

Share Image